

MAT 362 HW06

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Problem. 3.2.2. Show that if a surface is tangent to a plane along a curve, then the points of this curve are either parabolic or planar.

Solution. Suppose that the surface S is tangent to the plane P along the curve C . As the point p varies along C , $T_p(S) = P$ is constant, so $N(p)$ is also constant on C (since N being differentiable means that it can't flip to $-N(p)$). Therefore, if for a given $p \in C$, the tangent vector to C at p is w , $dN_p(w)$ must be equal to zero, hence dN_p isn't invertible and has determinant zero; thus the Gaussian curvature is zero at every point of C , which is exactly what it means for the points of the curve to be parabolic or planar. $\square$

Problem. 3.2.4. Assume a surface S has the property that $|k_1| \leq 1$, $|k_2| \leq 1$ everywhere. Is it true that the curvature k of a curve on S also satisfies $|k| \leq 1$?

Solution. $\boxed{\text{No}}$, $|k|$ may be greater than 1. Let S be the xy -plane, which satisfies $|k_1| = |k_2| = 0 \leq 1$ everywhere. Then, recall that the curvature of a circle with radius r is $\frac{1}{r}$ everywhere, so any circle in the xy -plane with radius less than 1 will have curvature larger than 1 at each point on the circle. $\square$

Problem. 3.2.5. Show that the mean curvature H at $p \in S$ is given by

$$H = \frac{1}{\pi} \int_0^\pi k_n(\theta) d\theta$$

where $k_n(\theta)$ is the normal curvature at p along a direction making an angle θ with a fixed direction.

Solution. Let e_1, e_2 be the principal directions at p . Letting $k_n(\theta)$ be the normal curvature at p along the direction making an angle of θ with e_1 , we get

$$\begin{aligned} \int_0^\pi k_n(\theta) d\theta &= \int_0^\pi II_p(e_1 \cos \theta + e_2 \sin \theta) d\theta \\ &= \int_0^\pi -\langle dN_p(e_1 \cos \theta + e_2 \sin \theta), e_1 \cos \theta + e_2 \sin \theta \rangle d\theta. \end{aligned}$$

Using the linearity of the inner product, we can write the integral as

$$\int_0^\pi II_p(e_1) \cos^2 \theta + II_p(e_2) \sin^2 \theta - \sin \theta \cos \theta \cdot (\langle dN_p(e_1), e_2 \rangle + \langle dN_p(e_2), e_1 \rangle) d\theta.$$

Then we can calculate

$$\int_0^\pi \cos^2 \theta d\theta = \int_0^\pi \frac{\cos(2\theta) + 1}{2} d\theta = \frac{\pi}{2} + \frac{1}{2} \int_0^\pi \cos(2\theta) d\theta = \frac{\pi}{2}$$

since

$$\int_0^\pi \cos(2\theta) d\theta = \frac{1}{2} \sin(2\pi) - \frac{1}{2} \sin(2 \cdot 0) = 0,$$

and then

$$\int_0^\pi \sin^2 \theta d\theta = \int_0^\pi 1 - \cos^2 \theta d\theta = \pi - \int_0^\pi \cos^2 \theta d\theta = \pi - \frac{\pi}{2} = \frac{\pi}{2}.$$

Additionally, by the identity $\sin(2\theta) = 2 \sin \theta \cos \theta$,

$$\int_0^\pi \sin \theta \cos \theta d\theta = \frac{1}{2} \int_0^\pi \sin(2\theta) d\theta = \frac{1}{2} \cdot -\frac{1}{2} (\cos(2 \cdot \pi) - \cos(2 \cdot 0)) = 0.$$

Therefore,

$$\frac{1}{\pi} \int_0^\pi II_p(e_1 \cos \theta + e_2 \sin \theta) d\theta = \frac{1}{\pi} \cdot \frac{\pi}{2} (II_p(e_1) + II_p(e_2)) = \frac{k_1 + k_2}{2}$$

as desired. $\square$

Problem. 3.2.8. Describe the region of the unit sphere covered by the image of the Gauss map of the following surfaces:

- (a) Paraboloid of revolution $z = x^2 + y^2$.
- (b) Hyperboloid of revolution $x^2 + y^2 - z^2 = 1$.
- (c) Catenoid $x^2 + y^2 = \cosh^2 z$.

Solution. Recall in general that if f is a differentiable function for which a is a regular value then the vector (f_x, f_y, f_z) is normal to the surface $f(x, y, z) = a$.

- (a) The paraboloid of revolution has an orientation given by

$$N(x, y, x^2 + y^2) = \frac{1}{\sqrt{4x^2 + 4y^2 + 1}}(2x, 2y, -1).$$

We will abbreviate this as $N(x, y)$ since the z -coordinate of a point on the surface is uniquely determined by x and y . We show that the image of this map is the hemisphere H of S^2 with negative z -coordinate. Clearly the image is a subset of H , so we need to verify that every element of H is in the image of N . That is, given $(a, b, c) \in S^2$ with $c < 0$, there exist real x, y such that

$$N(x, y) = \left(\frac{2x}{\sqrt{4x^2 + 4y^2 + 1}}, \frac{2y}{\sqrt{4x^2 + 4y^2 + 1}}, \frac{-1}{\sqrt{4x^2 + 4y^2 + 1}} \right) = (a, b, c).$$

We can check that $x = \frac{-a}{2c}$, $y = \frac{-b}{2c}$ works by first noting that

$$\sqrt{4x^2 + 4y^2 + 1} = \sqrt{4 \left(\frac{-a}{2c} \right)^2 + 4 \left(\frac{-b}{2c} \right)^2 + 1} = \sqrt{\frac{a^2 + b^2 + c^2}{c^2}} = \left| \frac{1}{c} \right| = \frac{-1}{c},$$

and then calculating

$$N(x, y) = -c \left(2 \cdot \frac{-a}{2c}, 2 \cdot \frac{-b}{2c}, -1 \right) = (a, b, c).$$

- (b) The hyperboloid of revolution has an orientation given by

$$N(x, y, z) = \frac{1}{\sqrt{x^2 + y^2 + z^2}}(x, y, -z).$$

We show that the image of N is the subset of S^2 strictly between the 45° and -45° latitudes (i.e. the set of points $(a, b, c) \in S^2$ such that $|c| < \sqrt{a^2 + b^2}$). Clearly the image of N is contained in this set because $x^2 + y^2 - z^2 = 1$ implies $z^2 < z^2 + 1 \leq x^2 + y^2$. Conversely, if $(a, b, c) \in S^2$ satisfies $|c| < \sqrt{a^2 + b^2}$, then we wish to find a solution to $N(x, y, z) = (a, b, c)$. Take

$$(x, y, z) = \left(\frac{a}{\sqrt{1 - 2c^2}}, \frac{b}{\sqrt{1 - 2c^2}}, \frac{-c}{\sqrt{1 - 2c^2}} \right)$$

(note that the denominators are real numbers because $c^2 < a^2 + b^2$ implies $c^2 < \frac{1}{2}$). This satisfies

$$x^2 + y^2 - z^2 = \frac{a^2 + b^2 - c^2}{1 - 2c^2} = \frac{1 - c^2 - c^2}{1 - 2c^2} = 1$$

so the point lies on the hyperboloid, and $N(x, y, z) = (a, b, c)$ is obvious because $N(x, y, z)$ is just the unit vector in the direction of $(x, y, -z) = \frac{1}{\sqrt{1 - 2c^2}}(a, b, c)$.

(c) The catenoid has an orientation given by

$$N(x, y, z) = \frac{1}{\sqrt{x^2 + y^2 + (\cosh z \sinh z)^2}}(x, y, -\cosh z \sinh z).$$

We show that the image of N is all of S^2 except the poles $(0, 0, \pm 1)$. Clearly $N(x, y, z)$ cannot equal $(0, 0, \pm 1)$ because this requires $x = y = 0$ but $\cosh^2 z \geq 1$ for all z . Conversely, I claim that

$$(x, y, z) = \left(\frac{a}{1 - c^2}, \frac{b}{1 - c^2}, \operatorname{arctanh}(-c) \right)$$

is a solution to $N(x, y, z) = (a, b, c)$ for all $(a, b, c) \in S^2$ with $|c| \neq 1$. First we can compute

$$-c = \tanh(z) = \frac{\sinh(z)}{\cosh(z)} = \frac{\sqrt{\cosh^2 z - 1}}{\cosh z}$$

which upon squaring and rearranging implies that $\cosh^2 z = \frac{1}{1 - c^2}$. This means that

$$x^2 + y^2 = \frac{a^2 + b^2}{(1 - c^2)^2} = \frac{1}{1 - c^2} = \cosh^2 z,$$

so (x, y, z) lies on the surface. Then,

$$\begin{aligned} N(x, y, z) &= \frac{1}{\sqrt{\cosh^2 z + \cosh^2 z \sinh^2 z}}(x, y, -\cosh z \sinh z) \\ &= \frac{1}{\cosh z \sqrt{1 + \sinh^2 z}}(x, y, -\cosh z \sinh z) \\ &= \frac{1}{\cosh^2 z}(x, y, -\cosh z \sinh z) \\ &= (x(1 - c^2), y(1 - c^2), -\tanh z) \\ &= (a, b, -(-c)) = (a, b, c) \end{aligned}$$

thus verifying that N is a surjective map from the catenoid to $S^2 \setminus \{(0, 0, 1), (0, 0, -1)\}$. $\square$